

Beyond the Keystones: What Physics Tells Us About Quantum Advantage That Complexity Theory Cannot

A Response to Huang, Choi, McClean & Preskill (2025)

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Abstract

Huang, Choi, McClean & Preskill (2025) have produced the most honest document about quantum advantage to emerge from the quantum computing establishment. But their five-keystone framework operates at the mathematical abstraction layer — qubits as Bloch sphere points, gates as unitary matrices, error correction as a code. It never descends into the physics of actual qubit implementations. This response identifies four physical keystones the framework overlooks: the self-referential circularity of quantum state calibration, a category error at the heart of the transmon qubit (calling a bosonic excitation a “photon” while using it as if Pauli exclusion applied), systematic vocabulary drift across four incompatible physics domains, and the gap between textbook qubits and laboratory qubits. I propose two additional keystones — Metrological Independence and Ontological Coherence — apply the extended framework to Google’s Sycamore experiment as a worked example, anticipate four counterarguments, and provide operational falsification conditions. The aim is not to refute Huang et al. but to extend their framework toward the physics it acknowledges it has not yet reached.

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1. Introduction

Huang, Choi, McClean & Preskill (2025) [1] have produced the most intellectually honest document about quantum advantage to emerge from the quantum computing establishment. Their five-keystone framework — Predictability, Typicality, Robustness, Verifiability, Usefulness — is the systematic evaluation

the field has needed for thirty years. Theorem 1 (detecting quantum advantage is classically hard, assuming $\text{BPP} \neq \text{BQP}$) is genuinely original. And their candor is remarkable: $\text{BPP} \neq \text{BQP}$ remains unproven; quantum recommendation systems were dequantized by Tang (2019); asymptotic sensing advantage is “fundamentally unattainable” in generic noise; and “we have never tested quantum mechanics at the complexity frontier where thousands of particles become massively entangled.”

But the framework operates entirely at the mathematical abstraction layer. Qubits are Bloch sphere points. Gates are unitary matrices. Error correction is a code. None of this descends into what actually happens inside a dilution refrigerator.

This response draws four arguments from the physics of real qubit implementations and proposes two new keystones. Section 6 applies the extended framework to Sycamore as a worked example. Section 7 anticipates objections. The aim is not to refute Huang et al. but to extend their framework toward the physics it acknowledges it has not yet reached.

2. The Self-Referential Metrology Problem

2.1 Calibration Is a Circle

Calibrating a transmon qubit means finding its resonant frequency ω , anharmonicity $\chi = \omega_{12} - \omega_{01}$, the pulse amplitude for a π -rotation, the readout phase response for $|0\rangle$ and $|1\rangle$, and cross-talk coupling. Every step depends on every other.

Multi-level spectroscopy probes the bare Hamiltonian’s eigenvalues without restricting to the two-level subspace — the closest we have to an independent anchor. But Rabi oscillations require knowing the readout calibration. Ramsey interferometry requires calibrated pulses. Gate set tomography [2] jointly estimates gates and measurements — guaranteeing internal consistency but not external accuracy, due to a gauge freedom: different gauge-equivalent gate sets produce exactly the same measurement statistics.

The calibration is a circle: calibrated gates define “known” states, calibrated readout measures them, outcomes update the calibration. Every element is defined in terms of every other.

2.2 The Spectroscopic Anchor

Spectroscopy of the bare Hamiltonian provides a partial anchor. Koch et al. (2007) [3] showed that anharmonicity can be measured by two-tone spectroscopy — this measurement does not assume the qubit subspace exists. It treats the transmon as the multi-level system it is.

The anchor is not perfect. The same readout chain — Josephson parametric amplifier, HEMT, digitizer — serves both spectroscopy and qubit readout. But the residual circularity is bounded by the readout fidelity, and that bound can be quantified.

2.3 When Circularity Becomes the Bottleneck

At today’s gate fidelities (99.9% single-qubit, 99% two-qubit), calibration circularity is a second-order concern. But the fault-tolerance threshold requires $\sim 10^{-4}$ per-gate error rates, and logical error rates of 10^{-15} demand physical error characterization at comparable precision. In this regime, we cannot independently verify our error models. Gate set tomography gives self-consistent estimates with gauge freedom. Randomized benchmarking [4] uses the same hardware. Spectroscopy anchors the computational subspace but does not validate the gate error model.

Google’s Sycamore [5] used cross-entropy benchmarking — self-consistent, not independently anchored. IBM’s Eagle [6] used error mitigation assuming accurately characterized noise. Neither reported an independent calibration uncertainty budget.

3. The Boson-Pauli Confusion

3.1 The Frequency-Domain Trick

A transmon qubit [3] is described as a “two-level quantum system.” It is not. It is a bosonic mode with an infinite ladder $|0\rangle, |1\rangle, |2\rangle, |3\rangle, \dots$ — a harmonic oscillator made anharmonic by the $\cos$ Josephson potential. The “two-level” restriction is maintained by **frequency selectivity**: a microwave pulse at cannot efficiently drive $|1\rangle \rightarrow |2\rangle$ because $\omega_{12} \neq \omega_{01}$.

This is engineering, not ontology. DRAG pulses are designed to cancel leakage to $|2\rangle$. Measurement-induced mixing with $|2\rangle$ is a known error channel. Thermal photons populate higher levels.

3.2 Boson = No Exclusion; “Two-Level” = Effective Exclusion

The transmon excitation is called a “photon” — borrowed from quantum optics, where photons are bosons that do not obey the Pauli exclusion principle. You can put arbitrarily many photons into a cavity mode.

But the transmon’s “two-level system” is treated as if exclusion applies: 0 or 1 photon. This exclusion is *engineered* — maintained by spectral isolation and active leakage suppression — not a consequence of particle statistics.

The category error is sharp: the word defined by the *absence* of exclusion describes a system whose entire engineering philosophy creates *effective* exclusion through frequency-domain tricks. The Josephson junction’s anharmonicity does the work Pauli exclusion would do for free in a fermionic system. But the isolation is imperfect, the gap is finite, and leakage is measurable.

3.3 The EJ/EC Trade-Off

Increasing EJ/EC exponentially suppresses charge noise — this is why the transmon exists. But residual anharmonicity *decreases* as EJ/EC increases. The more charge-noise-robust the qubit, the less anharmonic, the more harmonic-oscillator-like, and the harder to isolate the two-level subspace. This trade-off is bosonic at root.

Majorana zero modes in topological superconductors [7, 8, 9] are fermionic and genuinely obey Pauli exclusion. The protection is topological, not spectral. The question the Huang framework cannot ask — because it doesn’t descend to the physical layer — is whether the transmon’s frequency-domain isolation is fundamentally limited in ways fermionic architectures are not.

4. Domain Translation Errors

4.1 One System, Four Vocabularies

The vocabulary of a transmon processor is borrowed from incompatible domains. “Photon” (quantum optics: ~ 500 THz) means ~ 5 GHz in circuit QED. “Cavity” (Fabry-Pérot) means a coplanar waveguide on a chip [10]. “Plasmon” (electron density oscillation) means Cooper pair oscillation across a JJ. “Gate” (silicon logic) means a microwave pulse. Every term carries ontological baggage that does not always survive translation.

4.2 The Electron That Isn’t There

In a superconducting transmon, the Standard Model electron does not appear. The charge carriers are Cooper pairs — bosonic bound electron states, emergent quasiparticles. The “plasmon” excitation is their collective oscillation. Yet the vocabulary invites the image of something particle-like, and this image propagates into press releases and funding narratives.

Huang et al.’s acknowledgment that “we have never tested quantum mechanics at the complexity frontier” is the single most important sentence in their paper. But it does not trace this admission to its root: vocabulary inherited from low-entanglement particle physics is deployed to describe high-entanglement engineered many-body systems, and the semantic drift at the translation boundary is invisible to the mathematical framework evaluating it.

5. The Gap Between the Textbook and the Laboratory

5.1 Engineers Think Relationally

Ask a calibration engineer what they are doing. “I’m measuring anharmonicity, finding the Rabi frequency for a 20 ns π -rotation, optimizing the DRAG parameter.” Frequency-domain addressability, modal occupation numbers, spectral isolation. No billiard balls.

The particle-qubit picture lives in popular science, quantum computing textbooks for computer scientists [11], and investor presentations. The confusion is pedagogical and financial — not operational.

5.2 Two Quantum Technologies

There are two “quantum technologies” at play:

- **Pedagogical:** Abstract qubits as two-level systems, unitary gates as matrices, error correction as a code. Scale up $\rightarrow$ threshold theorem guarantees success.
- **Operational:** Transmon chips with frequency-domain tricks, calibration circles, correlated errors, amplifier saturation. Scale up $\rightarrow$ problems invisible at single-qubit level become dominant.

The Huang framework evaluates the pedagogical technology. The operational technology contains its own keystones the framework cannot see. The gap between them is the single largest unquantified uncertainty in quantum computing.

6. Extending the Framework

6.1 Two Additional Keystones

#	Keystone	Definition	How to Evaluate
1–5	Predictability, Typicality, Robustness, Verifiability, Usefulness	Per Huang et al. (2025)	Per Huang et al. (2025)

#	Keystone	Definition	How to Evaluate
6	Metrological Independence	Calibration uncertainty bounded by independent anchors, quantified and reported	Specify spectroscopic anchor, GST gauge freedom, residual uncertainty budget
7	Ontological Coherence	Vocabulary identifies physical entities without cross-domain category errors	Specify physical implementation, degree of freedom, isolation mechanism, readout chain demarcation

Keystone 6 asks: what independent anchors were used, what residual calibration uncertainty remains, and is it smaller than reported error rates by a sufficient factor?

Keystone 7 asks: what physical implementation is the “qubit”? Where in the amplifier chain does projection occur? Is the vocabulary faithful to the physics, or does it borrow from incompatible domains?

6.2 Worked Example: Sycamore (2019)

Google’s 53-qubit Sycamore experiment [5] claimed quantum supremacy: a random circuit sampling task in 200 seconds that would take a classical supercomputer 10,000 years. How does the extended framework evaluate this?

Keystone	Assessment	Evidence
Predictability	PARTIAL — conditional on BPP BQP	Founding conjecture unproven 30 years
Typicality	PASS	Random circuits are worst-case for classical simulation
Robustness	PARTIAL — demonstrated at 53 qubits	Error rates too high for fault tolerance at this scale; threshold theorem guarantees future scalability but with unknown overhead

Keystone	Assessment	Evidence
Verifiability	PARTIAL — gap too large to cross-verify	10,000-year gap precludes classical verification; subsequent classical algorithms reduced the gap but did not eliminate it
Usefulness	FAIL	Random circuit sampling has no known practical application
Metrological Independence	FAIL	No independent calibration uncertainty budget reported; cross-entropy benchmarking is self-consistent, not independently anchored
Ontological Coherence	FAIL	Platform described as “53 superconducting qubits” without identifying the physical degree of freedom or isolation mechanism; fidelity characterization and solver map assumed on same platform

Under the extended framework, Sycamore passes 1 keystone (Typicality), partially passes 3 (Predictability, Robustness, Verifiability), and fails 3 (Usefulness, Metrological Independence, Ontological Coherence). The original five-keystone framework gives Sycamore a mixed but defensible score. The extended seven-keystone framework is stricter — not because the physics refutes the experiment, but because the experiment’s reporting does not include the physical metadata the keystones demand.

This does not mean Sycamore was invalid. It means its scientific reporting was incomplete under a framework that requires physical grounding.

6.3 Falsification Conditions

C1 — Metrology. If any platform demonstrates gate fidelities below fault-tolerance threshold with independent calibration error budgets verified to the same precision without GST circularity, §2 is refuted.

Verification: Monitor published calibration procedures.

C2 — Hardware. If a fault-tolerant quantum computer performs commercially useful computation at lower joules-per-solution than any classical alternative before 2035 using exclusively Archimedean implementations, the combined argument is refuted.

Verification: Monitor benchmarks; track joules-per-solution.

C3 — Vocabulary. If a structured terminology audit across 100+ quantum computing papers (2019–2026) finds borrowed terms carry domain-of-origin semantics faithfully with no systematic drift — inter-rater reliability ≥ 0.8 — then §4 is refuted.

Verification: Independent coders classify usage patterns.

7. Objections and Responses

I anticipate four counterarguments and address each.

7.1 “Calibration circularity is a solved problem — gate set tomography handles it.”

Gate set tomography provides self-consistent estimates, not externally verified ones. The gauge freedom means different gauge-equivalent gate sets produce identical measurement statistics but assign errors to different parts of the circuit. For a decoder, it matters whether errors are attributed to single-qubit gates, two-qubit gates, or measurements. GST alone cannot resolve this ambiguity — which is why randomized benchmarking and multi-level spectroscopy are used as partial cross-checks.

The point is not that calibration is impossible. It is that calibration uncertainty is not currently reported, and at the fault-tolerance threshold, unreported uncertainty becomes the bottleneck. This is a demand for transparency, not a claim of impossibility.

7.2 “The Boson-Pauli point is semantic, not substantive — engineers know the transmon is anharmonic.”

Precisely the opposite. Engineers know the physics. The category error is in how the field *communicates* advantage claims to funders, journalists, and textbook readers based on a mathematical model that engineers already recognize as a scaffold.

When a press release claims “53 superconducting qubits performed a computation in 200 seconds that would take 10,000 years classically,” the listener imagines 53 quantum particles doing something particle-like. The engineer knows it is 53 frequency-domain isolation zones in a bosonic circuit. The gap between

these two descriptions is not cosmetic — it is the mechanism by which overclaiming propagates from the laboratory to the marketplace. This is an operational concern, not a semantic one.

7.3 “You haven’t proven that calibration circularity limits fault tolerance — you’ve asserted it.”

Correct. That is why §6.3 provides falsification conditions. If a fault-tolerant device is demonstrated meeting the conditions of C1 and C2, the argument of this response is refuted. The burden of proof should fall on those claiming quantum advantage, not on those questioning it. Huang et al. would agree: they explicitly state that advantage claims require “rigorous, substantive, and quantifiable evidence.”

The further point is that the required evidence has a physical dimension that the current framework does not capture. Metrological Independence makes that dimension explicit.

7.4 “What do you propose instead of the transmon architecture?”

Three paths, none exclusive:

1. **Fermionic qubits.** Majorana zero modes in topological superconductors [7, 8, 9] offer genuine fermionic exclusion — no frequency-domain trick needed. The engineering challenges are substantial, but the physical principle is different from the transmon, and it bypasses the EJ/EC trade-off entirely.
2. **Non-Archimedean approaches.** Ostrowski’s theorem [12] shows that $\mathbb{Q}$ has exactly two types of completions: Archimedean $\mathbb{R}$ and p-adic $\mathbb{Q}_p$. p-adic quantum mechanics [13, 14] extends quantum theory to non-Archimedean fields. Whether this translates into physical qubit protection is an open hypothesis — not a proven framework — but it is a hypothesis the current framework cannot evaluate because it does not distinguish topologies.
3. **The software-first path.** p-adic discrete gate compilation — encoding quantum logic on non-Archimedean gate sets while running on existing transmon or trapped-ion hardware — is a software layer testable on current cloud quantum computers. This is the most pragmatic test: if it improves fidelity, the hypothesis gains support; if not, it is weakened. Either outcome advances the field.

The broader response is that proposing alternatives is not a prerequisite for identifying a gap. The Huang framework identifies gaps in advantage claims without proposing alternative algorithms. My argument identifies gaps in the framework itself — and where possible, sketches what might fill them.

8. Conclusion

Huang, Choi, McClean & Preskill have produced a landmark document. Their framework and candor set a standard the rest of the field should emulate.

But the framework is incomplete at the physical layer. I have argued that the physical implementation of qubits introduces keystones invisible to complexity theory: calibration circularity that scales with system size, a category error encoded in the word “photon” itself, vocabulary drift across four incompatible physics domains, and a gap between textbook qubits and laboratory qubits that few advantage claims bridge.

The Sycamore worked example (§6.2) demonstrates that applying the extended framework is not abstract philosophy — it changes how we evaluate a specific, widely cited advantage claim. The objections section (§7) shows that the arguments withstand anticipated counterarguments: calibration is not a solved but an unreported problem, the Boson-Pauli point is operational not semantic, the burden of proof is correctly placed, and alternatives exist even if they are not prerequisites.

The paper I most want to read applies this extended framework, with all seven keystones, to every major quantum advantage claim of the past decade — Sycamore, Eagle, Willow, and whatever comes next — asking for each: what is the independent calibration anchor, what is the residual uncertainty, and what fundamental physical trade-offs does the implementation impose?

The most honest document about quantum advantage is, by its own admission, incomplete. That incompleteness is not a weakness. It is an invitation. This response is an attempt to accept it.

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